COMPLETE SYSTEM ARCHITECTURE & FEATURE GUIDE

# Dastarkhwan AI

Urdu & Roman Urdu Conversational Voice Ordering & Real-Time Kitchen Operations Platform for Pakistani Restaurants

**Project Type:** Final Year Project (FYP) & Enterprise Architecture Prototype

**Frontend Technologies:** React 19, Vite 8, Tailwind CSS, Web Speech API, Web Audio API

**Backend Foundation:** Laravel 12 API Architecture Ready

**Generated Date:** October 2026

**Status:** Fully Interactive Prototype (Live on Localhost)

# 1. Executive Summary & Problem-Solution Fit

## 1.1 Real-World Pakistani Food Ordering Bottlenecks

The quick-service restaurant (QSR) and fast-food industry in Pakistan experiences severe operational challenges during lunch and dinner rushes that standard Western software suites cannot solve:

- **Extreme Peak-Hour Telephone Congestion:** Between 8:00 PM and 11:30 PM, restaurant phone lines are continuously engaged. Customers face 5–10 minute hold times, causing a 30–40% abandonment rate.
- **Colloquial Language & Slang (Code-Switching):** Customers speak in a seamless blend of Roman Urdu and English (e.g., "*Bhai 2 zinger burger aur aik thanda pepsi bhej do*", "*nahi 3 nahi 2 kar do*", "*extra spicy sauce daalna*"). Traditional English voicebots fail to parse these phonetic nuances.
- **Cash-on-Delivery (COD) Refusal & Fake Order Losses:** Over 85% of restaurant deliveries in Pakistan are settled in cash at the doorstep. Restaurants suffer substantial food waste and dispatch costs when customers refuse delivery or submit fake orders.
- **Kitchen Miscommunication & Lost Tickets:** Telephone order-takers write paper slips by hand or manually enter items into the POS, leading to missing modifiers, incorrect quantities, and delayed dispatches.
- **Selling 86'd (Out-of-Stock) Dishes:** When key kitchen inventory runs out during a rush, telephone agents continue taking orders for those dishes due to a lack of real-time communication.
- **Customer Frustration & Churn:** Conventional bots get trapped in repetitive loops without providing a way to reach a human manager.

## 1.2 How Dastarkhwan AI Automates the Operations

| Traditional Problem | Dastarkhwan AI Automated Solution |
| --- | --- |
| **Busy Phone Lines & Hold Times** | **0ms Automated Concurrency:** AI Voice and Chat agents process multiple orders simultaneously 24/7 without hold times. |
| **Roman Urdu & Slang Misunderstandings** | **Custom Keyword & Entity Parser:** Grounds customer intent against menu aliases and recognizes Roman Urdu numerals ( `do` , `teen` , `ek` , `char` ), Urdu script ( `دو` , `ایک` ), and English words. |
| **Wrong Items & Customer Misunderstandings** | **Automatic Bill Read-Back:** The AI summarizes the itemized bill, delivery charges, and address, requiring explicit confirmation ( `"Haan" / "Ji sahi hai"` ) before cooking starts. |
| **Fake Orders & COD Fraud** | **Explainable Heuristic Risk Scorer:** Automatically evaluates order risk (0–100) based on past cancellations, basket value, and customer history. High-risk orders trigger an automated verification phone call. |
| **Delayed Kitchen Punching** | **Instant Live Kanban Dispatch:** Confirmed orders appear instantly in the kitchen's "New" column with a Web Audio API two-tone bell chime and visual highlight animation. |
| **Selling Sold-Out (86'd) Items** | **Real-Time 86'd Stock Sync:** Toggling an item to "Off" in the admin dashboard instantly informs all voice and chat agents to reject that item and suggest available alternatives. |
| **Angry Customer Churn** | **Sentiment Analysis & Human Handoff:** Detects negative sentiment keywords ( `bakwas` , `ghussa` , `fuzool` ) or repeated misunderstandings and routes the conversation to a Store Manager console. |

# 2. Complete UI Breakdown & Feature Directory

Every individual interface, page, modal, and interactive control in the Dastarkhwan AI platform is itemized below:

## 2.1 Global Top Bar & Navigation Controls

### 🧡 Brand & Environment Tag

Displays the Dastarkhwan AI branding, platform title, and `FYP Prototype` tag.

### 👥 Demo Persona Selector

Dropdown to switch between 3 customer profiles:

- *Ayesha Khan (New Customer):* First-time user, triggers standard new-account evaluation.
- *Zainab Ahmed (Regular VIP):* 14 successful past orders, 0 cancellations, evaluates as Low Risk.
- *Ali Raza (Flagged Account):* 4 previous delivery cancellations, triggers High Risk rules.

### ▶️ "Demo Mode" Walkthrough Button

Launches the automated 5-step interactive scripted presentation showing Roman Urdu parsing, corrections, and kitchen dispatch.

### 🔄 "Reset Demo" State Button

Instantly restores default mock orders, clears the cart, resets chat history, clears manager escalations, and resets stock levels.

## 2.2 Persistent Sidebar Navigation

- **Order Menu ( `/menu` ):** Food catalog with add-to-cart controls and stock status indicators.
- **AI Chat Ordering ( `/chat` ):** Smartphone simulator with live NLP intent inspector.
- **Kitchen Kanban ( `/dashboard` ):** Full-width 4-column kitchen fulfillment board with pending order counter badge.
- **Human Handoff ( `/handoff` ):** Real-time escalation queue accompanied by an amber pulsing alert badge.
- **Risk & Fraud Check ( `/risk` ):** COD fraud mitigation console with red alert counter badge.
- **Kitchen Analytics ( `/analytics` ):** Order volume, hourly peak rushes, and top-seller charts.
- **Future Features ( `/roadmap` ):** Strategic enterprise expansion roadmap (telephony, WhatsApp, digital wallets).
- **AI Engine Status Indicator:** Live green indicator confirming: "*AI Engine Online — Urdu / Roman Urdu Keyword Parser Grounded on Menu*".

## 2.3 Page 1: Order Menu Hub ( `/menu` )

- **Dual-View Switch (Customer View vs Admin Mode):** Toggles between customer shopping and kitchen stock management.
- **Stock Summary Strip:** Live counter displaying the total count of `In Stock` vs `Sold Out (86'd)` items.
- **Category Filter Bar:** Quick-filter pills: 🌟 *All Items*, 🍔 *Burgers*, 🍕 *Pizzas*, 🍟 *Sides*, 🥤 *Drinks*, and 🔥 *Value Deals*.
- **Fuzzy Search Input:** Search box matching English names, Urdu script titles, and phonetic aliases (e.g., `zingar`, `fajita`, `thanda`).
- **Visual Menu Item Card:**
  - High-resolution food photography for all 21 items.
  - Overlaid price pill in PKR (e.g., `Rs. 520`).
  - Best Value combo badge with flame icon.
  - Urdu script subtitle (e.g., `کرسپی زنگر برگر`).
  - Item description and ingredients.
  - **"Add to Cart" Button (Customer View):** One-tap button with animated green checkmark confirmation ( `"Added to Cart!"` ).
  - **86'd Toggle Button (Admin View):** One-tap button to mark dishes as *In Stock* or *Sold Out*, instantly updating AI agent knowledge.
- **Slide-Out Cart Drawer:**
  - Itemized cart list with unit prices.
  - Quantity adjusters ( `+` and `-` ) with real-time recalculation of subtotal and standard delivery fee ( `Rs. 150` ).
  - Item deletion trash button.
  - **"Place Order" Button:** Directly dispatches the order to the kitchen Kanban board with an audio chime.
  - **"Clear Cart" Button:** Empties the cart.
- **Mobile Floating Cart Bar:** Bottom shortcut bar appearing on mobile screens when items are in the cart.

## 2.4 Page 2: Customer AI Chat Ordering ( `/chat` )

- **Smartphone Frame Mockup:** Realistic bezel with camera notch, Wi-Fi/battery status bar, and active customer profile header.
- **"Talk to a human" Emergency Button:** Visible button allowing customers to immediately request a store manager.
- **Message Feed:** Distinct message bubbles for the Customer (orange), AI Assistant (slate), and Store Manager (highlighted).
- **Animated Typing Indicator:** Three bouncing dots shown while the AI evaluates customer speech.
- **Quick-Reply Chips:** Horizontal scrollable suggestion chips: `Menu dikhao` 🍔, `Deals kya hain?` 🔥, `2 zinger de do`, `Order confirm`, `Talk to Human`.
- **Input Bar:** Text field supporting English, Roman Urdu, and Urdu script, along with an inline microphone button.
- **"What the AI Understood" NLP Inspector Card:**
  - Displays detected linguistic style ( `Roman Urdu`, `Urdu Script`, or `English` ).
  - Displays verbatim customer input text.
  - Verifies menu grounding status (confirming zero hallucinations).
  - Displays parsed items, detected quantities, and extracted modifiers (e.g., `Extra Spicy`, `Chilled` ).
- **Side Cart & Fast Checkout Card:** Displays active cart items and subtotal alongside the phone mockup.

## 2.5 Persistent Floating AI Voice & Chat Assistant

*Accessible from the bottom-right corner across all pages via the floating action button (FAB).*

- **Floating Action Button (FAB):** Fixed at `bottom-6 right-6` with pulse ring animations when listening, unread notification dots, and a real-time cart item count badge.
- **Expandable 560px Assistant Window:**
  - **Header Controls:** Online status indicator, animated audio waveform, voice synthesizer mute toggle ( `Volume2 / VolumeX` ), and close button.
  - **Tab 1: Text Chat:** Full conversational chat synced with global state, quick chips, and message history.
  - **Tab 2: Voice Order:**
    - **Language Selector:** Dropdown to switch between `English / Roman Urdu (en-US)` and `اردو (ur-PK)`.
    - **Interactive Microphone Orb:** 96px central button with state transitions (Idle, Listening with red pulse rings, Speaking with green pulse rings).
    - **Live Speech Transcript Box:** Real-time speech-to-text transcript feed.

        - **Barge-in Interrupt Button:** "Stop Speaking" button to immediately halt TTS speech synthesis when the user speaks.
        - **Agent Voice Response Box:** Displays the text of the AI's spoken answer.
    - **Bottom Mini Cart Strip:** Shows item count, total price, and a direct "Place Order" button.

## 2.6 Page 3: Kitchen Kanban Dashboard ( `/dashboard` )

- **Operational Stats Strip:** Displays Total Sales Today (PKR), Orders Dispatched, Average Order Value (AOV), and Pending Orders.
- **4-Column Kitchen Kanban Board:**
    1. **New Orders:** Incoming orders arrive here with an automated Web Audio API two-tone bell chime ( `587.33Hz -> 880Hz` ) and a 4.5-second emerald border glow.
    2. **Kitchen Preparing:** Orders actively being cooked.
    3. **Ready for Dispatch:** Cooked orders waiting for delivery riders.
    4. **Completed / Delivered:** Successfully fulfilled orders.
- **Interactive Kanban Order Card:**
    - Order ID (e.g., `#PK-102` ), customer name, phone number, and channel tag ( `Voice Agent` , `Chat Agent` , `Menu` ).
    - Delivery address with map pin icon.
    - Time elapsed counter (e.g., `Just now` , `12 mins ago` ).
    - Itemized dish list with quantities and modifiers (e.g., `Extra Spicy` , `Chilled Pepsi` ).
    - Order grand total formatted in PKR.
    - **Fraud Risk Badge:** Color-coded `Low` , `Medium` , or `High` risk indicator.
    - **"Confirmation Call Needed" Button:** Appears on high-risk orders to launch the phone verification modal.
    - **Status Advance Controls:** `Move to Preparing` , `Mark Ready` , `Delivered` .
    - **Cancel Order Button:** Cancels the order and updates revenue totals.
    - **"Transcript" Button:** Opens the complete conversational transcript modal for that order.

## 2.7 Page 4: Human Handoff & Attention Desk ( `/handoff` )

- **Customer Escalation Queue:** Lists all customer conversations requiring human intervention with customer name, phone number, and timestamp.
- **Escalation Triggers:**
  - *Customer Frustration / Anger:* Triggered by sentiment analysis ( `bakwas` , `ghussa` , `fuzool` , `fraud` ).
  - *Manager Request:* Triggered by phrases like `talk to human` or `manager se baat karni hai` .
  - *Double Comprehension Failure:* Triggered when the AI fails to parse intent twice consecutively.
- **Manager Takeover Console:** Displays the full chat transcript and allows the store manager to send messages with an official `[Store Manager]` tag.
- **"Resolve & Return to AI" Button:** Closes the escalation and removes it from the attention queue.
- **"Simulate Escalation" Button:** Test button to simulate a customer escalation on demand.

## 2.8 Page 5: Fake-Order Risk & Verification ( `/risk` )

- **Risk Filter Buttons:** Filters orders by `All` , `High Risk` , `Medium` , and `Low` .
- **Explainable Fraud Heuristic Rules:**
  - *Past Delivery Refusals:* 3+ cancellations adds +45 risk points; 1–2 adds +25 points.
  - *First-Time Customer:* Unverified address adds +15 points.
  - *High COD Basket Value:* Orders over Rs. 3,000 add +15 points; over Rs. 5,000 add +30 points.
  - *Suspicious Item Quantities:* Single item qty ≥ 4 or total items ≥ 7 adds +15 points.
- **Order Verification Card:** Displays numerical risk score bar (0–100), itemized breakdown reasons, and action buttons ( `Call Customer` , `Verify Order` , `Reject & Cancel` ).

## 2.9 Page 6: Kitchen Analytics & Performance ( `/analytics` )

- **KPI Cards:** Total Sales Today, Orders Dispatched, Average Order Value (AOV), and AI Automation Rate (94.2%).
- **Hourly Peak Rush Chart:** Visual bar chart of order volume across Pakistani meal times (Lunch Rush, Afternoon Slump, Dinner Start, Peak Rush at 9 PM, Late Night, Midnight Cravings).
- **Best Sellers Leaderboard:** Top 5 revenue-generating dishes with order counts, PKR revenues, and progress bars.
- **Channel Breakdown:** Volume split between Voice Agent, Chat Agent, and Web Menu.

## 2.10 Page 7: Future Enterprise Roadmap ( `/roadmap` )

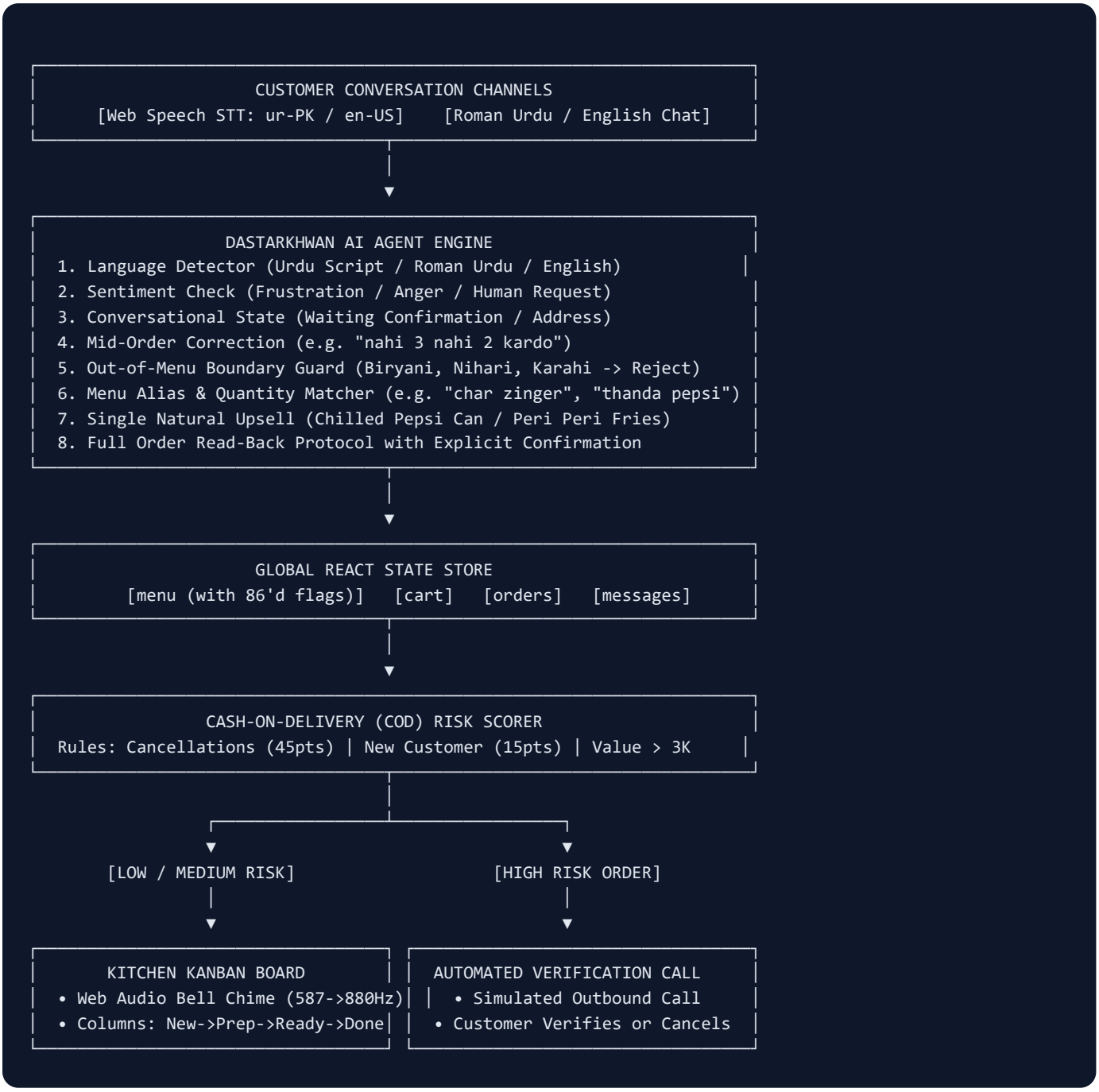
- **Phase 6: Real Telephony & SIP Trunking:** Inbound call ingestion via Twilio/Exotel with Deepgram Urdu STT and ElevenLabs voice generation.
- **Phase 7: Official WhatsApp Cloud API Bot:** Ordering on WhatsApp with interactive catalog menus and GPS location pin-drop address capture.
- **Phase 8: Pakistani Digital Payments (JazzCash, Easypaisa, Raast):** Instant mobile wallet prompts and Raast QR scanning to eliminate doorstep cash refusal.
- **Phase 9: Kitchen Demand Forecasting & Wastage AI:** Predictive algorithms estimating ingredient usage and peak rushes.
- **Phase 10: Multi-Branch Routing & POS Synchronization:** Radius-based kitchen auto-dispatch with bidirectional sync into Pakistani POS systems (Foodnerd, Tossdown, Retailer POS).

## 2.11 Interactive Modals

- **Automated COD Confirmation Call Modal ( `ConfirmationCallModal` ):** Simulates an automated telephone call to verify high-risk COD orders with multi-stage dialer (Dialing → Ringing → Connected), live call timer, and customer voice dialogue.
- **Demo Mode Auto-Play Walkthrough Modal ( `DemoModeModal` ):** Automated 5-step presentation demonstrating Roman Urdu parsing, mid-conversation corrections, out-of-menu rejection, checkout read-back, and live kitchen dispatch.
- **Order Details & Full Transcript Modal ( `OrderDetailsModal` ):** Audit view showing complete customer details, delivery address, and the verbatim chat/voice transcript for that order.

# 3. Core Engine & Technical Architecture

## 3.1 System Flow Diagram



## 3.2 Technical Service Specifications

| Service Module | File Path | Primary Technical Responsibilities |
|---|---|---|
| **Global Store** | `src/context/AppContext.jsx` | Manages `menu`, `cart`, `orders`, `chatMessages`, `attentionQueue`, and generates the pure Web Audio API bell chime without external audio files. |
| **Conversational Engine** | `src/services/agentEngine.js` | Rule-based NLP engine with Pakistani numeral parsing ( `ek`, `do`, `teen`, `دو`, `ایک` ), alias matching ( `zingar`, `thanda` ), out-of-menu guardrails, one-time upsell logic, and sentiment analysis. |
| **Speech Service** | `src/services/speechService.js` | Web Speech API wrapper handling Speech-to-Text ( `ur-PK` / `en-US` ), Text-to-Speech synthesis with emoji stripping, and instant **barge-in interruption** when the user speaks. |
| **Fraud Risk Scorer** | `src/utils/riskScorer.js` | Explainable heuristic scoring engine evaluating COD fraud risk (past cancellations, first-time status, basket value, bulk quantities). |
| **Food Catalog** | `src/data/menu.js` | 21 Pakistani restaurant items across 5 categories with high-resolution imagery, English names, Urdu script titles, prices, and phonetic aliases. |
| **Customer Profiles** | `src/data/customers.js` | Realistic Pakistani customer personas representing new, regular VIP, and high-cancellation users. |

# 4. End-to-End Verification and Testing Guide

Follow these steps to test every workflow in your browser (Google Chrome or Microsoft Edge recommended):

1. **Visual Menu & Slide-Out Cart:**
   - Navigate to `http://localhost:5173/menu`.
   - Click category filter pills and search for `zinger`.
   - Click **Add to Cart** on a Zinger Burger and a Large Pizza.
   - Open the **Cart Drawer**, adjust quantities using `+` and `-`, and click **Place Order**.
   - Verify the order appears in `Kitchen Kanban` (`/dashboard`) under the `New` column.
2. **Admin Stock 86'd Real-Time Sync:**
   - On `/menu`, toggle to **Admin Mode**.
   - Mark `Crispy Zinger Burger` as **Mark 86'd**.
   - Open the floating AI assistant at the bottom right and type: `1 zinger burger de do`.
   - Verify the AI informs you that Zinger Burger is sold out and suggests an alternative.
3. **Voice Ordering with Barge-In:**
   - Open the floating assistant and switch to the **Voice Order** tab.
   - Tap the microphone orb and speak: `2 fajita pizza aur do thandi pepsi`.
   - Observe the live transcript, the audio response, and the updated cart total.
   - Speak while the agent is responding to verify barge-in immediately silences the speech synthesis.
4. **Cash-on-Delivery Risk & Verification Call:**
   - In the Top Bar, change the Persona to **Ali Raza (Flagged Account)**.
   - Place a large order (> Rs. 3,500) via chat or menu.
   - Navigate to `Risk & Fraud Check` (`/risk`).
   - Observe the `High Risk` badge and the itemized risk reasons.
   - Click **Call Customer** to observe the telephone dialer, ring, and connected audio flow.
5. **Human Handoff & Store Manager Console:**
   - In the chat interface, type: `Bhai bakwas service hai, manager se baat karni hai`.
   - Observe the AI immediately transferring you to the manager.
   - Navigate to `Human Handoff` (`/handoff`) to view the escalated conversation.
   - Send a reply as the store manager and verify it appears in the customer's phone view with a `[Store Manager]` tag.

6. **Automated Scripted Demo Walkthrough:**
    - Click **Demo Mode** in the Top Bar and click **Play**.
    - Watch the scripted 5-step ordering flow demonstrate Roman Urdu handling, quantity correction, out-of-menu rejection, and final kitchen dispatch.